

| Assessment | Test | Domain        | Skill                  | Difficulty                               |
|------------|------|---------------|------------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | Hard <div></div> <div></div> <div></div> |

Question

$$(9x^4 + 8x^3) - (7x^3 + 5x^2) - (3x^2 + x)$$

The given expression can be written in the form  $9x^4 + x^3 + ax^2 - x$ , where  $a$  is a constant. What is the value of  $a$ ?

Answer

- A.  $-8$
- B.  $-2$
- C.  $2$
- D.  $8$

Correct Answer: A

Rationale

Choice A is correct. The given expression is equivalent to  $9x^4 + 8x^3 - 7x^3 - 5x^2 - 3x^2 - x$ . By combining like terms, this can be written as  $9x^4 + x^3 - 8x^2 - x$ . This expression is in the form  $9x^4 + x^3 + ax^2 - x$ , where  $a$  is a constant. Therefore, the value of  $a$  is  $-8$ .

Choice B is incorrect. This may result from adding  $-5x^2$  and  $3x^2$  rather than  $-5x^2$  and  $-3x^2$ .

Choice C is incorrect. This may result from adding  $5x^2$  and  $-3x^2$  rather than  $-5x^2$  and  $-3x^2$ .

Choice D is incorrect. This may result from adding  $5x^2$  and  $3x^2$  rather than  $-5x^2$  and  $-3x^2$ .